A light wave is incident from air on a thick glass slab of refractive index 1.8, with angle of incidence 30°. Determine the fraction of light power reflected from, and transmitted through, the air–glass interface. Assume TE polarization. What should the incidence angle be to transmit 95% of the power through this interface.

Calculate the thickness of a Ge crystal layer needed for absorbing 90% of incident radiation at 1550 nm. Repeat for $In_{0.53}Ga_{0.47}As$.

Consider a 15-mW laser at wavelength 532 nm. Calculate the photon flux of this laser.

A laser beam is incident perpendicular to the surface of one of the short faces of a 45–45–90 glass prism. If the refractive index of glass is 1.5, show that the light undergoes total internal reflection when it strikes the long face of the prism. Explain all steps with a sketch and label it.

Find the transmitted power ratio between the red sun light striking sea water normally, and that of violet sun light. Which will penetrate deeper into water? Explain. For water: $n(red) = 1.3314 - j3.48 \times 10^{-8},\ n(violet) = 1.34451 - j2.11 \times 10^{-10}$.

A ray of red light in air is incident on the surface of an unknown non-magnetic dielectric material at an angle of 39°. It was found that the refracted ray makes an angle of 27°. Find the refractive index of this material

The average solar photon flux density is about 10^{14} photons/(s.cm^2). Assume that the average solar radiation wavelength is 0.5 μm, and calculate light power received by a 1-m radius circular solar cell panel.

Calculate the reflected power ratio at air-GaAs interface at normal incidence between red (680 nm) and blue (470 nm) lights.

With a certain metal plate, shining a beam of red light on the metal causes electrons to be emitted. (a) If we replace the red light by blue light, do we know that electrons will be emitted? (b) If the two beams have the same intensity and are incident on equal areas of the plate, do we get the same number of electrons emitted per second in the two cases? (c) The plate material is most probably Cs (W=1.85 eV), Na (W=2.4 eV) or Ca (W=2.9 eV). Choose one and explain.

Unpolarized light is incident on a smooth flat floor (n=2.2) at an angle of 45°. What is the polarization of the reflected light?